Poison-resilient continual intrusion detection on CICDDoS2019

Can a gate plus a frozen teacher keep a replay-based NIDS safe AND still learning when its stored labels are poisoned?

Main result (four labels, CII, seeds 42-44)
With ALL 2,000 stored Syn labels flipped to BENIGN, teacher relabelling learns all four classes in 3/3 seeds on held-out test:
Syn 0.9952 · LDAP 0.9825 · UDP 0.9983 · benign FAR 0.0011 (gate only and soft repair: LDAP = UDP = 0)

HELD-OUT TEST POST HOC the relabel method was chosen after the pre-registered sweep's diagnostic

- **Scope:** four labels defended (BENIGN, Syn, LDAP, UDP); one attacked class (Syn); CII; 3 seeds; no significance claim
- **Not claimed:** a 12- or 14-label defence (unsolved), or training on all 70.4M raw rows (we trained on capped, deduplicated subsets)
- **Known limitation:** a recall-based gate does not see confidence loss (RF study, slide 10)

Source: github.com/DevChi1l/cicddos2019-continual-nids (committed files; tables: docs/RESULT_TABLES.md)

Problem, threat model and what is new here

- **Continual NIDS:** a detector keeps learning new attack types; replay mixes stored old rows into each update so it does not forget
- **Attack surface:** whoever can edit stored labels can make it forget an old attack or block new learning
- **Threat model:** flip a share f of the stored labels of ONE old class (Syn -> BENIGN). The attacker cannot touch new data, validation, test, the teacher or gradients
- **Inherited:** CI/CII scenarios, replay + buffer label-flipping threat, joint oracle (reference paper); feature-token Transformer (FT-Transformer, Gorishniy et al. 2021); CICDDoS2019 (Sharafaldin et al. 2019)

Our contribution (implemented and measured)

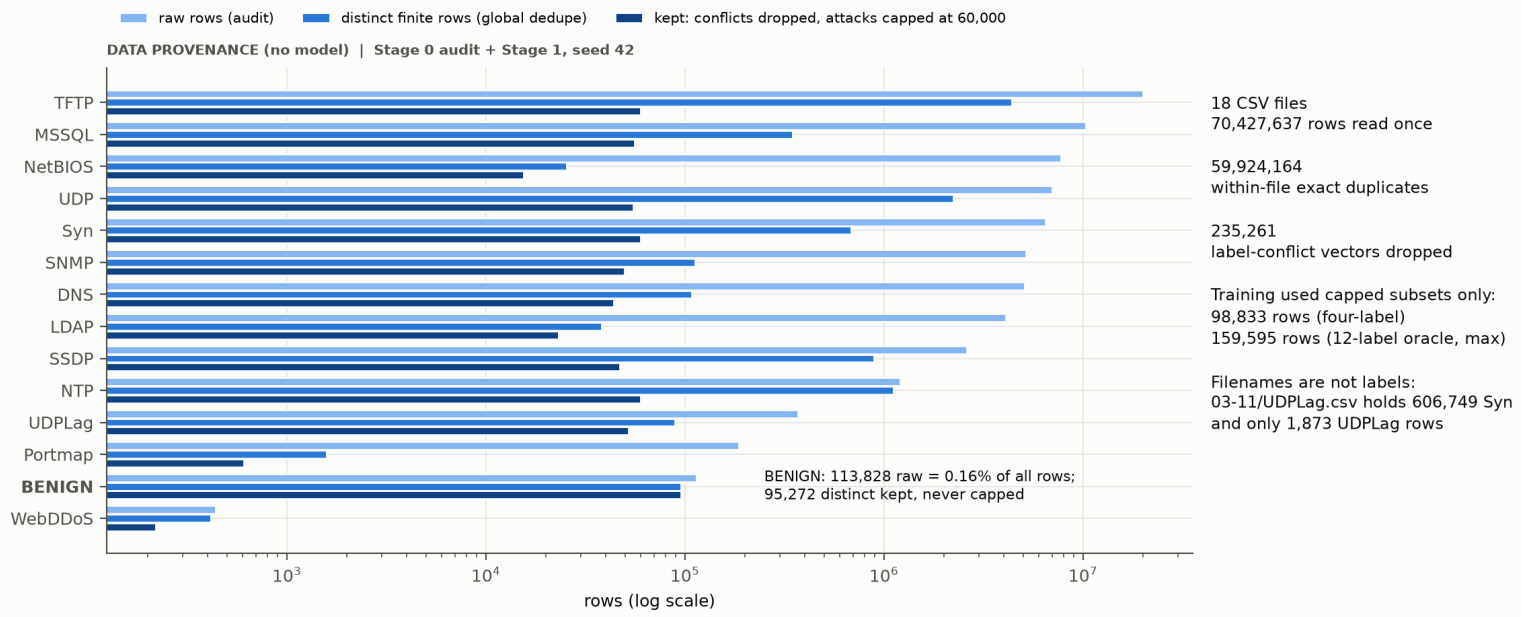
- Leakage-controlled pipeline over all 70.4M raw rows (audit, global dedupe, conflict drop, E0-only scaler, transformed-duplicate exclusion)
- Validation gate + frozen E0 teacher + one repair attempt; rejected update -> keep last accepted model
- Pre-registered poison sweep; post-hoc teacher relabelling (FU2) as the defence at 100% flips
- Honest limits: 12-label protocol fails; two pilots stopped; confidence loss invisible to the gate

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Audited dataset and preprocessing

DATA

1. From 70.4M raw rows to deduplicated, capped classes (14 raw labels)



Source: notebooks/manifest.json; experiments/results/01_data/split_manifest.json

- **Audit:** 18 CSV files, 70,427,637 rows, 14 raw labels, 59,924,164 within-file duplicates
- **Order:** label column (not filename) -> drop ids, IPs, ports, timestamp, Inbound -> drop non-finite -> global dedupe on a feature hash -> drop 235,261 label-conflict vectors -> cap 60,000 distinct per attack -> split 60/10/10/20
- **BENIGN:** 113,828 raw (0.16%) -> 95,272 distinct kept, never capped
- **Inputs:** 77 numeric features + protocol
- **Trained on:** capped subsets only (98,833 rows four-label; at most 159,595)

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How E0 / E1 / E2 are formed (four-label, CII)

experience	new class	attack rows	fresh BENIGN rows	model after it
E0	Syn	13,890	19,055	T (frozen) and A0
E1	LDAP	13,890	19,054	A1 (if accepted)
E2	UDP	13,890	19,054	A2 (if accepted)

- **Splits fixed first:** validation 23,263 rows, held-out test 46,527 rows (19,028 benign); never used for training
- **Scaler:** signed log1p -> standardise -> clip ±5, fitted on E0 training rows only, then frozen
- **Leakage guard:** 166 protected rows excluded because their scaled input equals a training input (residual 0)
- **Replay buffer:** 2,000 rows per class per ACCEPTED experience, true label stored; a rejected experience adds nothing

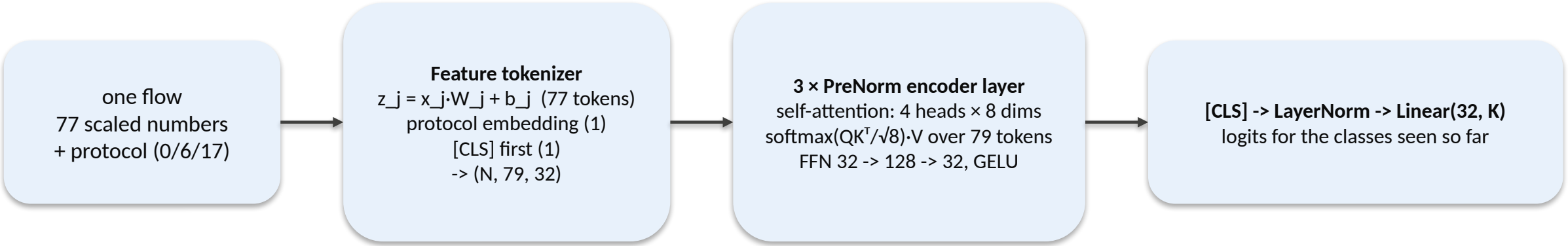
Attack cap = 13,890
= LDAP's whole cleaned training count, so every attack class contributes equally

Same Stage 1 splits feed every experiment (sha256 bb1f5acb…)

12-label runs: 9,312 per class, two classes per update; RF: fixed 3 classes

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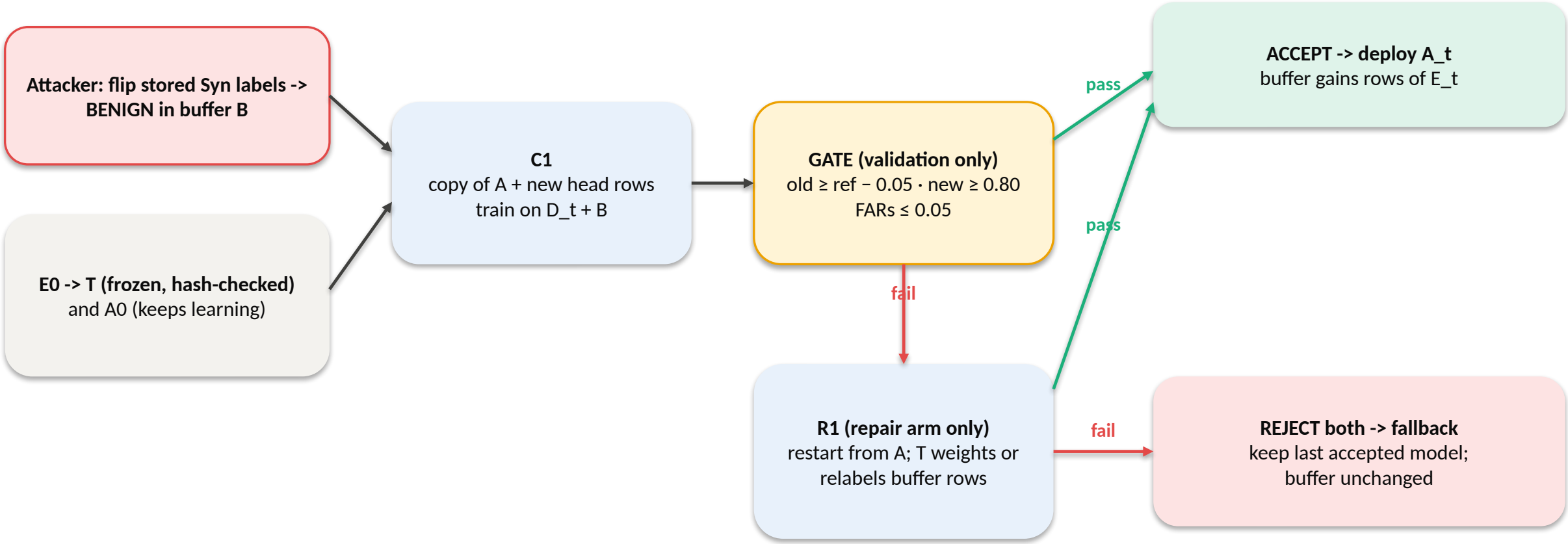
Model: FT-Transformer-style feature-token Transformer



- **Size:** d = 32; 43,396 parameters with 4 classes (43,660 with 12)
- **New classes:** the head grows; old output rows are copied, so old logits are unchanged before training
- **Loss:** class-weighted cross-entropy, c_k = sqrt(N / (K·n_k)), × a per-row weight (used only by repair); AdamW, 10 epochs for E0, 5 per update
- **Closest published design:** FT-Transformer's feature tokenizer + [CLS] + PreNorm (smaller; GELU not ReGLU). Not the original TabTransformer

Source: github.com/DevChi1l/cicddos2019-continual-nids (committed files; tables: docs/RESULT_TABLES.md)

Replay poisoning and the defended update protocol



- **Soft R1 (pre-registered):** row weight = p_T(stored label | x). Relabel R1 (FU2, post hoc): if p_T(stored) ≤ 0.1 and T is ≥ 0.9 sure of another class, train the row with T's label
- **T is never trained after E0** and knows only BENIGN and Syn; test rows are never read by any decision

Source: github.com/DevChi1l/cicddos2019-continual-nids (committed files; tables: docs/RESULT_TABLES.md)

Why relabelling: the sweep, then FU2's validation decisions

VALIDATION POST HOC

f = 1.0: all 2,000 stored Syn labels flipped; seeds 42-44

seed	update	attempt	Syn val recall	Syn floor	rows relabelled (to true class)	gate
42	E1	C1 (stored labels)	0.0000	0.9453	-	REJECT
42	**E1**	**R1 relabel**	**0.9953**	**0.9453**	**1995 (1995)**	**ACCEPT**
42	E2	C1 (stored labels)	0.0000	0.9453	-	REJECT
42	**E2**	**R1 relabel**	**0.9953**	**0.9453**	**1995 (1995)**	**ACCEPT**
43	E1	C1 (stored labels)	0.0000	0.9461	-	REJECT
43	**E1**	**R1 relabel**	**0.9941**	**0.9461**	**1994 (1994)**	**ACCEPT**
43	E2	C1 (stored labels)	0.0000	0.9441	-	REJECT
43	**E2**	**R1 relabel**	**0.9955**	**0.9441**	**1994 (1994)**	**ACCEPT**
44	E1	C1 (stored labels)	0.0000	0.9461	-	REJECT
44	**E1**	**R1 relabel**	**0.9961**	**0.9461**	**1995 (1995)**	**ACCEPT**
44	E2	C1 (stored labels)	0.0000	0.9461	-	REJECT
44	**E2**	**R1 relabel**	**0.9960**	**0.9461**	**1995 (1995)**	**ACCEPT**

Pre-registered sweep (context)

PRE-REGISTERED HELD-OUT TEST

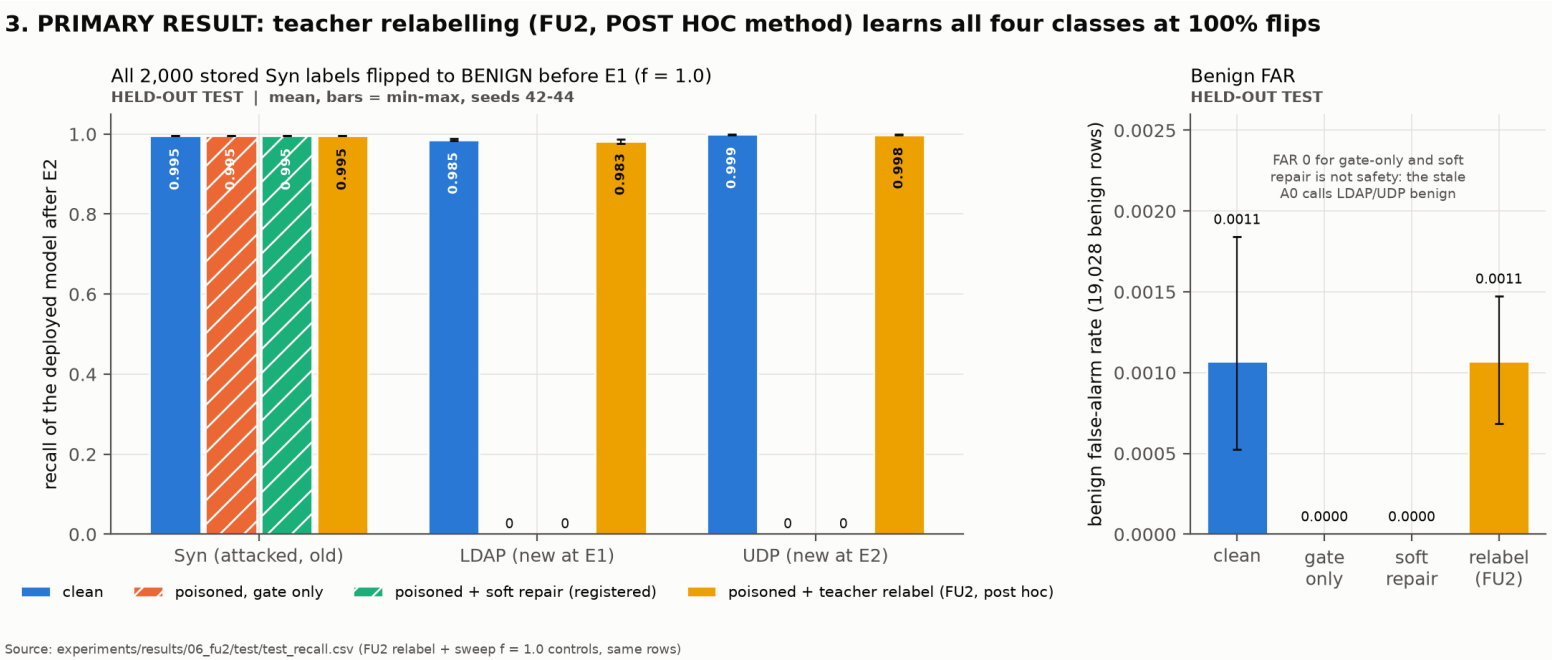
- f ≤ 0.95: all 4 classes learned in 9/9 soft-repair runs vs 3/9 gate-only
- f = 1.0: soft R1 accepted 0/6; teacher weight on flipped rows ≈ 0.003: wrong labels removed, no Syn label left
- Relabel diagnostic would have passed 3/3 -> FU2 deploys it (registered before its run)

Source: github.com/DevChi1l/cicddos2019-continual-nids (committed files; tables: docs/RESULT_TABLES.md)

Result: four-class defence at 100% flips (held-out test)

HELD-OUT TEST | POST HOC

mean (min-max) over seeds 42-44; 46,527 test rows, 19,028 benign



arm	Syn	LDAP	UDP	benign FAR
clean	0.9954	0.9851	0.9988	0.0011
gate only	0.9953	0.0000	0.0000	0.0000
soft repair	0.9953	0.0000	0.0000	0.0000
relabel (FU2)	**0.9952**	**0.9825**	**0.9983**	**0.0011**

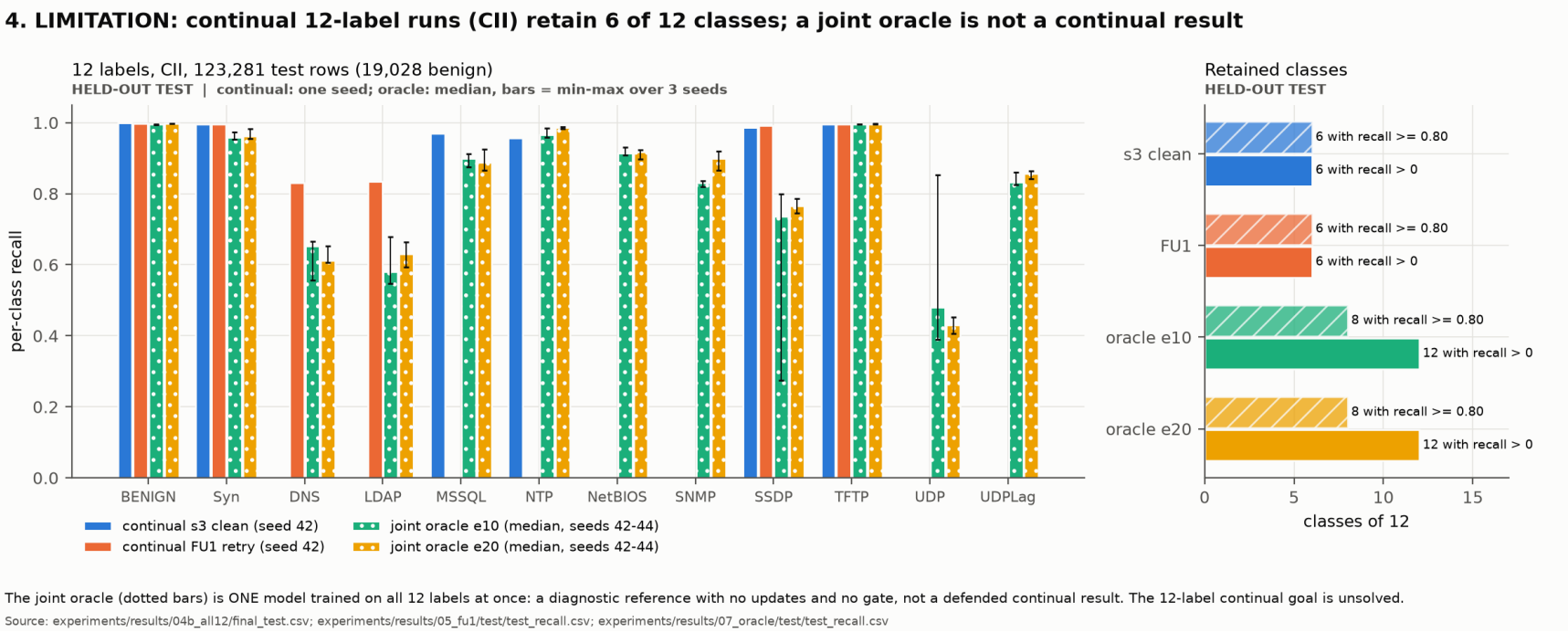
- H3 (Syn ≥ 0.95, LDAP & UDP ≥ 0.80, FAR ≤ 0.05) met in 3/3 seeds
- FAR 0 for gate-only is NOT safety: the stale A0 labels LDAP/UDP traffic benign
- Scope: 4 labels, Syn attacked, CII, 3 seeds

Source: github.com/DevChi1l/cicddos2019-continual-nids (committed files; tables: docs/RESULT_TABLES.md)

Limitation: the 12-label continual setting is not solved

HELD-OUT TEST | LIMITATION | DIAGNOSTIC

oracle = one jointly trained model, not continual



The joint oracle (dotted bars) is ONE model trained on all 12 labels at once: a diagnostic reference with no updates and no gate, not a defended continual result. The 12-label continual goal is unsolved.

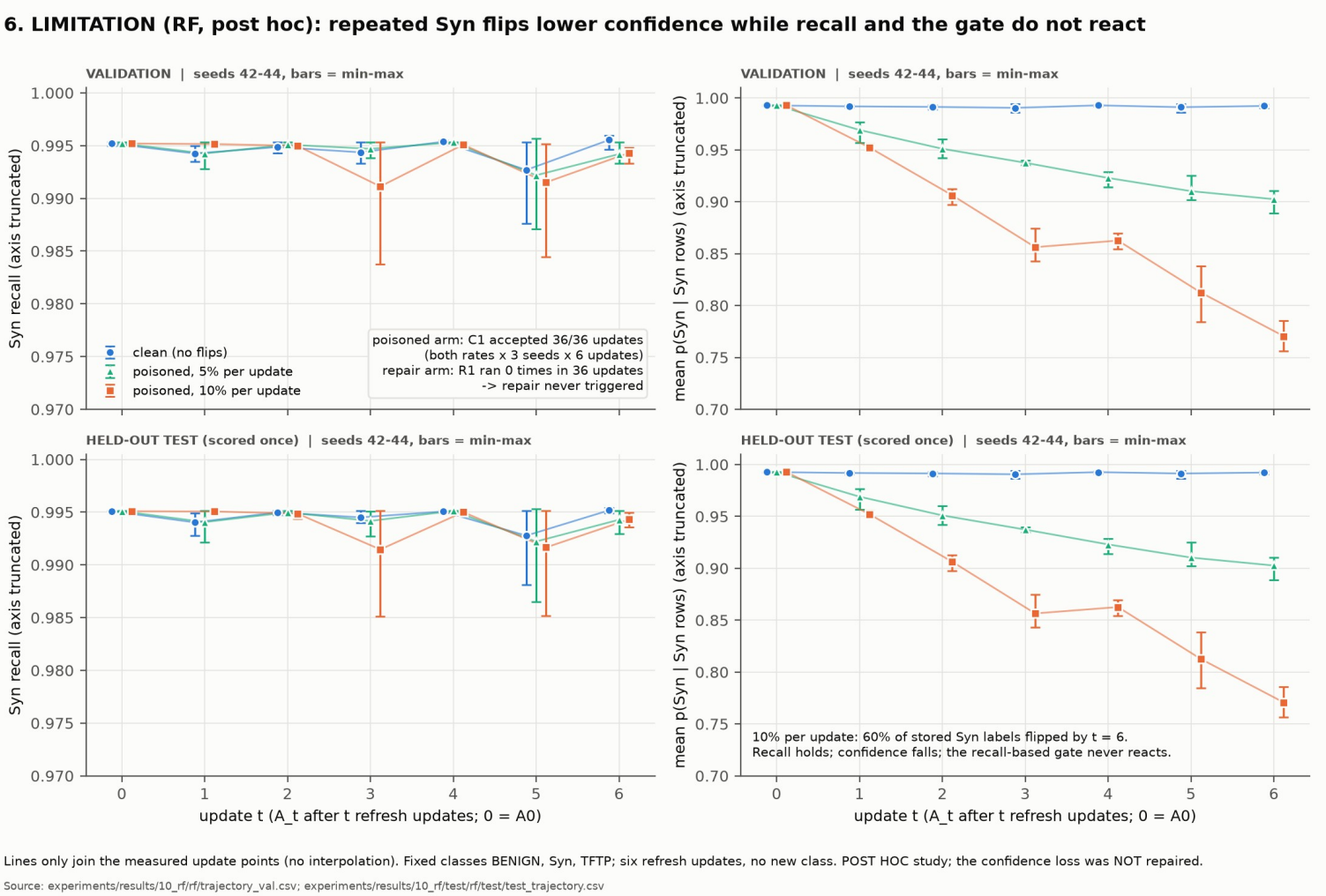
Source: experiments/results/04b_all12/final_test.csv; experiments/results/05_fu1/test/test_recall.csv; experiments/results/07_oracle/test/test_recall.csv

- **Continual s3 clean:** 6/12 classes, macro recall 0.492 (one seed)
- **FU1 retry:** 6/12, macro 0.471; DNS/LDAP 0.83/0.83 learned, MSSQL/NTP lost
- **Joint oracle (diagnostic):** macro 0.816/0.829; DNS 0.65, LDAP 0.63, UDP 0.48
- **Claim:** no 12- or 14-label defence

Source: github.com/DevChi1l/cicddos2019-continual-nids (committed files; tables: docs/RESULT_TABLES.md)

Limitation: confidence loss the recall-based gate does not see (RF)

VALIDATION | HELD-OUT TEST | POST HOC | LIMITATION



- **Setting:** BENIGN, Syn, TFTP fixed; 6 refresh updates; no new class
- **Attack:** +10% stored Syn labels flipped per update -> 60% after E6
- **Gate:** poisoned C1 accepted 36/36; repair R1 ran 0 times
- **Test, A6 (10%):** Syn recall 0.9943 vs 0.9952 clean; mean Syn confidence 0.7707 vs 0.9922
- **Not claimed:** that repair fixed this (it never ran); new-class arrival
- **Next (proposal):** a pre-registered confidence check in the gate

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